

Tiny Ultrasound Thermometry Demo Paper

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Abstract

Thermal ablation procedures require reliable temperature monitoring because excess heating can damage surrounding tissue. This tiny teaching paper summarizes how ultrasound thermometry can support evidence-grounded RAG demos.

Methods

A compact literature-style note was written with page-level metadata. The demo parser extracts text from each PDF page, detects section names, cleans whitespace, and keeps DOI, year, page, and section metadata for citations.

Discussion

Ultrasound thermometry may estimate temperature-related changes using speed of sound, attenuation, signal amplitude, and backscattered energy. The same parsed text can be chunked into retrieval units for later vector search.

Limitations

This PDF is synthetic and intentionally small. It demonstrates ingestion and debugging behavior, not clinical validity or a complete literature review.